

Answer to question 1

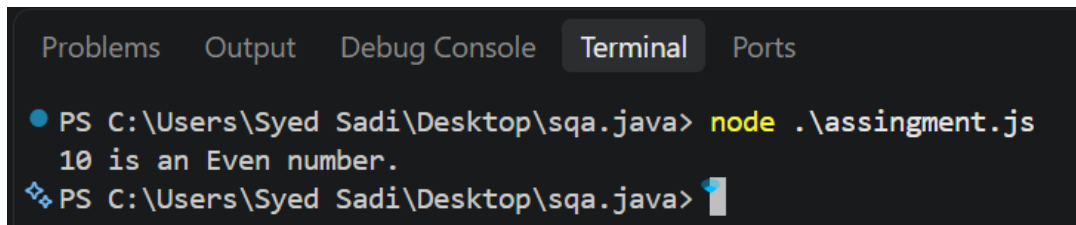
1. Problem statement: Check whether a number is even or odd

Source code :

```
const number = 10;
```

```
if (number % 2 == 0) {  
    console.log(number + " is an Even number.");  
} else {  
    console.log(number + " is an Odd number.");  
}
```

Output :



The screenshot shows a VS Code interface with a terminal window open. The terminal has tabs for 'Problems', 'Output', 'Debug Console', 'Terminal', and 'Ports'. The 'Terminal' tab is active. The command prompt shows the user running 'node .\assingment.js' in the directory 'C:\Users\Syed Sadi\Desktop\sqa.java'. The output of the command is '10 is an Even number.'.

```
Problems  Output  Debug Console  Terminal  Ports  
● PS C:\Users\Syed Sadi\Desktop\sqa.java> node .\assingment.js  
10 is an Even number.  
❖ PS C:\Users\Syed Sadi\Desktop\sqa.java>
```

Explanation:

The % (modulus) operator finds the remainder after division.

If a number divided by 2 has a remainder of 0 that will be an even number.

Otherwise that will be odd number.

2. Problem statement: Find the largest of three numbers

Source code :

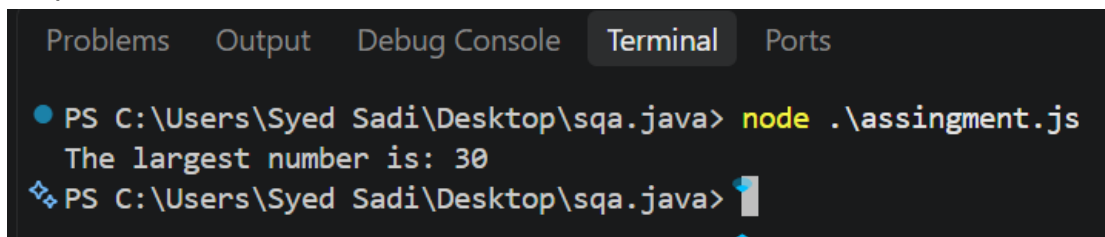
```
const num1 = 15;
const num2 = 30;
const num3 = 25;

let largest;

if (num1 >= num2 && num1 >= num3) {
    largest = num1;
} else if (num2 >= num1 && num2 >= num3) {
    largest = num2;
} else {
    largest = num3;
}

console.log("The largest number is: " + largest);
```

Output:



The screenshot shows a terminal window with tabs for 'Problems', 'Output', 'Debug Console', 'Terminal', and 'Ports'. The 'Terminal' tab is active. It displays a command prompt where the user has run 'node .\assingment.js' (note the typo in the filename). The output of the script is 'The largest number is: 30'. The prompt then shows the user's next command, 'PS C:\Users\Syed Sadi\Desktop\sqa.java>', with a cursor at the end.

```
PS C:\Users\Syed Sadi\Desktop\sqa.java> node .\assingment.js
The largest number is: 30
PS C:\Users\Syed Sadi\Desktop\sqa.java>
```

Explanation:

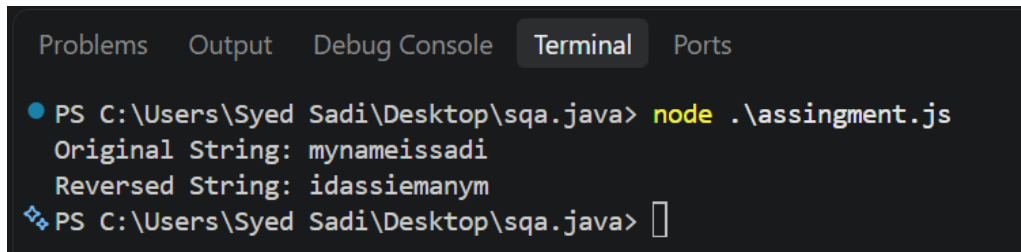
The code compares all three numbers using if, else if, and else conditions.

3. Problem statement: Reverse a string

Source code:

```
const text = "mynameissadi";  
const reversedText = text.split("").reverse().join("");  
  
console.log("Original String: " + text);  
console.log("Reversed String: " + reversedText);
```

Output:



The screenshot shows a VS Code terminal window with the 'Terminal' tab selected. The command prompt is 'PS C:\Users\Syed Sadi\Desktop\sqa.java>'. The user has run the command 'node .\assingment.js'. The output of the script is displayed as two lines: 'Original String: mynameissadi' and 'Reversed String: idassiemanym'. The prompt is now 'PS C:\Users\Syed Sadi\Desktop\sqa.java>' with a cursor.

Explanation:

Here,

split("") → Converts the string into an array of characters.

reverse() → Reverses the array.

join("") → Combines the characters back into a string.

4. Problem statement: Count vowels in a string.

Source Code:

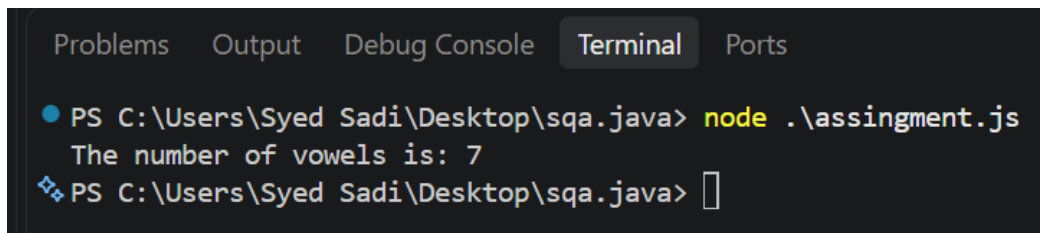
```
let count = 0;

for (let i = 0; i < text.length; i++) {
  const character = text[i].toLowerCase();

  if (
    character === "a" ||
    character === "e" ||
    character === "i" ||
    character === "o" ||
    character === "u"
  ) {
    count++;
  }
}

console.log("The number of vowels is: " + count);
```

Output:



The screenshot shows a terminal window with tabs for 'Problems', 'Output', 'Debug Console', 'Terminal', and 'Ports'. The 'Terminal' tab is active. It displays a command prompt where the command 'node .\assingment.js' has been executed. The output of the command is 'The number of vowels is: 7'. Below the output, the prompt is ready for the next command.

```
PS C:\Users\Syed Sadi\Desktop\sqa.java> node .\assingment.js
The number of vowels is: 7
PS C:\Users\Syed Sadi\Desktop\sqa.java> 
```

Explanation:

The program checks every character in the string using a for loop and the vowel counter increases by 1.

5. Problem statement: Remove duplicate values from an array

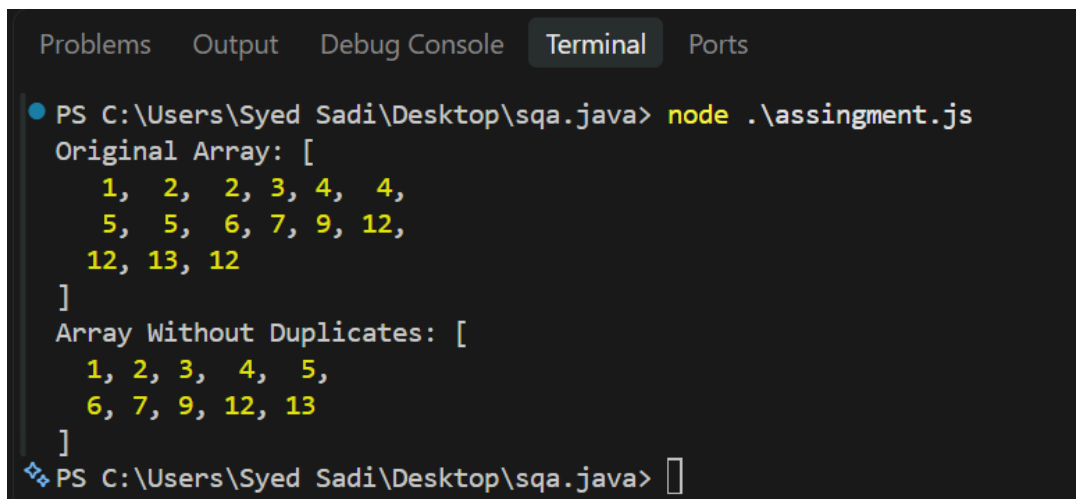
Source code :

```
const numbers = [1, 2, 2, 3, 4, 4, 5, 5, 6, 7, 9, 12, 12, 13, 12];

const uniqueNumbers = [...new Set(numbers)];

console.log("Original Array:", numbers);
console.log("Array Without Duplicates:", uniqueNumbers);
```

Output:



```
Problems  Output  Debug Console  Terminal  Ports

• PS C:\Users\Syed Sadi\Desktop\sqa.java> node .\assingment.js
Original Array: [
  1, 2, 2, 3, 4, 4,
  5, 5, 6, 7, 9, 12,
  12, 13, 12
]
Array Without Duplicates: [
  1, 2, 3, 4, 5,
  6, 7, 9, 12, 13
]
❖ PS C:\Users\Syed Sadi\Desktop\sqa.java> 
```

Explanation :

I used a Set object that stores unique values only.

Answer to question 2

1. What is Node.js?

Ans: Node.js is an open-source, cross-platform JavaScript runtime environment that allows developers to run JavaScript code outside a web browser.

Traditionally, JavaScript was mainly used for creating interactive web pages inside browsers. Node.js lets us use JavaScript for Server-side development, Backend applications, APIs, Automation scripts, Command-line, tools, Testing and QA automation.

2. How does Node.js work?

Ans: Node.js works by using a single-threaded, event-driven, non-blocking I/O model to execute JavaScript on the server side.

Its basic working process is:

- The Node.js application executes JavaScript code using the V8 Engine.
- Node.js uses an Event Loop to manage multiple tasks efficiently.
- When Node.js performs operations such as reading files, accessing databases, or making network requests, it does not necessarily wait for one operation to finish before handling other tasks.

3. Advantages of Node.js for QA Automation

Ans: Using Node.js for QA automation has become highly popular because it aligns perfectly with modern agile development.

The advantages are :

- Uses the V8 engine for fast execution.
- Can efficiently handle many simultaneous requests.
- Performs multiple operations without waiting.
- Uses the same language for frontend and backend.
- Provides millions of useful packages and tools.
- Works on Windows, macOS, and Linux.
- Suitable for chat, gaming, and live applications.
- Supported by a large global developer community.
- Frameworks like Express.js and NestJS simplify development.
- Lightweight and efficient for modern applications.

4. Difference between Browser JavaScript and Node.js

Feature	Browser JavaScript	Node.js
Environment	Runs inside a web browser	Runs outside the browser
Main Purpose	Creates interactive web pages	Builds backend applications and automation
DOM Access	Yes	No
window Object	Available	Not available
document Object	Available	Not available
File System Access	Limited for security reasons	Available through Node.js modules
Server Development	Not directly	Yes
npm Support	Indirectly used in development	Fully supported
QA Automation	Can support browser-based scripts	Widely used for automation tools